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A-to-I RNA editing in *Xanthomonas oryzae* pv. *oryzicola* enhances its pathogenicity and adaptation

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Adenosine-to-inosine (A-to-I) RNA editing is an important posttranscriptional event in eukaryotes; however, has been relatively unexplored in prokaryotes. This study investigated A-to-I RNA editing in *Xanthomonas oryzae* pv. *oryzicola* (Xoc) and discovered two editing events. The first event, induced by oxidative stress, resulted in an A-to-I change that recoded serine to proline (S128P) in the mRNA of *fliC*, encoding a flagellar filament protein. Using genome-wide RNA immunoprecipitation-coupled high-throughput sequencing (iRIP-Seq) analysis with HA-tagged TadA from Xoc, we found that TadA binds to the mRNA of *fliC*, and the binding motif is identical to that previously reported. The editing event led to changes in flagellar filament structure, increasing motility and enhancing tolerance to oxidative stress, as modelled in 3D and measured by TEM. The S128P mutant also showed an increase in biofilm formation, measured by 3D laser scanning confocal microscopy. RNA-seq analysis revealed that a gene cluster contributing to siderophore biosynthesis and Fe³⁺ uptake was upregulated in S128P compared with wild-type. Intracellular levels of reactive oxygen species and an oxidative stress survival assay indicated that this gene cluster reduces the Fenton reaction, increasing biofilm formation and bacterial virulence. Another A-to-I RNA editing event shows that *xfeA* in Xoc senses extracytoplasmic iron and changes the hydrogen bonding network of ligand channel domains. The frequency of A-to-I RNA editing during iron-deficient conditions increased by 76.87%, which facilitated the passage of iron through the XfeA outer membrane channel. When bacteria were subjected to high iron concentrations, the percentage of A-to-I editing in *xfeA* decreased, which reduced iron transport via XfeA. Furthermore, A-to-I RNA editing increased expression of multiple genes in the chemotaxis pathway, including methyl-accepting chemotaxis proteins (MCPs) that sense concentrations of exogenous ferrienterobactin (Fe-Ent) at the cytoplasmic membrane. In summary, our study demonstrates the importance of A-to-I RNA editing in bacterial pathogenicity and adaptation to environmental stress. These findings provide new insights into the mechanisms by which Xoc respond to extracellular iron and oxidative stress, which may lead to the development of novel targets for the control of bacterial leaf streak.